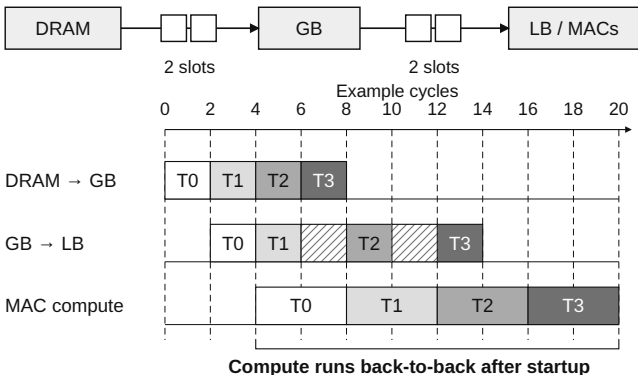


(b) Cycle-level timing and energy

Hardware determines elapsed time

Access latency, link bandwidth, resource availability

Illustrative pipeline: two buffer slots per boundary



 Stall: downstream buffer still occupied

T0–T3: work tiles; 2 / 2 / 4 cycles per stage.

Slot released after its consumer finishes.

Energy from the same traced activities

Access / operation counts × unit costs + leakage × time